

FUNCTIONAL SAFETY LIDAR • GS SERIES

TYPE 3 RATED OLEI GS1-5 SAFETY LIDAR

Your platform needs a 270° scanner that stops for people and can be signed off. The GS1-5 is that scanner — Type 3 per IEC 61496, SIL 2, Cat. 3, PL d, with two OSSD groups and 64 configurable protective regions — at a fraction of the price of the incumbents.



■ WHY THIS SENSOR

- **Type 3 (IEC 61496), SIL 2, Cat. 3, PL d** — the safety class a machine-safety engineer can put a signature against
- **Two OSSD output groups plus CANopen Safety** — wires into the platform's safety function directly, no interposing logic
- **64 protective regions and 128 secure regions** — up to two monitoring cases, switched live as the platform changes speed or route
- **≤ 5 m protective field at 2% reflectivity** — sized on the hard case — dark clothing, not a white test board
- **≤ 70 ms fastest response time, 30 ms scan cycle** — short enough to keep stopping distances honest at AMR speeds
- **≤ 40 m measuring range, 0.167° angular resolution** — warning fields well beyond the protective envelope
- **Configurable 20–150 mm protective-field resolution** — 20 mm minimum detectable object diameter for finger and leg detection
- **Colour LCD and LED status on the housing** — a technician can read the state of the device without a laptop

TYPE 3

IEC 61496

PL d / SIL 2

ISO 13849 • IEC 61508

≤ 70 ms

RESPONSE TIME

270°

SCAN ANGLE

■ APPLICATION SCENARIOS



PERSONNEL SAFETY

Rated protective fields around people and machines



AGV & AMR SAFETY

Rated stop and warn fields on driverless platforms



INDUSTRIAL AUTOMATION

Guarded cells, robot envelopes and access ways



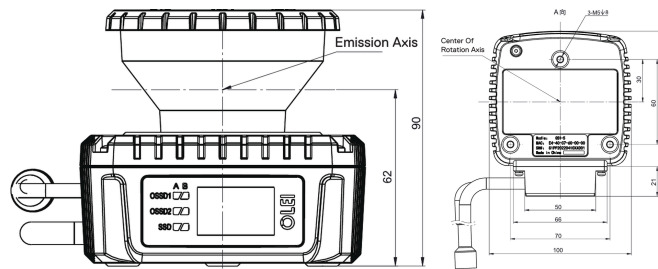
ANTI-PINCH

Platform-edge and door-line pedestrian protection

PARAMETER	TECHNICAL DATA
Type	Type 3 (IEC 61496)
Safety integrity level	SIL 2 (IEC 61508 / IEC 62061)
Category	Cat. 3 (ISO 13849)
Performance level	PL d (ISO 13849)
Scan angle	270°
Scope of protection	≤ 5 m @ 2% reflectivity
Measuring range	≤ 40 m @ 80% reflectivity
Number of protected regions	64 groups
Number of secure regions	128 units
Monitoring cases	≤ 2
Accuracy	Typical ±30 mm
Protection area resolution (configurable)	20 / 30 / 40 mm 50 / 75 / 150 mm
Angular resolution	0.167°
Response time	Fastest response time ≤ 70 ms
Multisampling	2-16
Scan cycle time	30 ms
Protective field supplement (generally necessary)	65 mm
Detectable reflectance	1.8% – high reflectivity
Light type	Pulsed laser diode
Laser safety level	Class 1
Laser wavelength	905 nm

PARAMETER	TECHNICAL DATA
Communication mode	CANopen Safety · Ethernet
CAN output	1 group
OSSD	2 groups
Configuration method	Safety config tool
Display device	Colour LCD screen, LED light
Protection class	III
Rated voltage	24 V DC (–30% to +20%)
Power consumption	≤ 6 W (no load)
Mechanical dimensions	115 × 100 × 90 mm
Weight	850 g
Enclosure protection class	IP65
Material	Aluminium case Window cover: PC
Connecting cables	LAN cable 0.3 m I/O & power cable 2 m
Installation method	Upright, reverse and side mounting Optional bracket
Operating / storage temperature	–10 to +50 °C Storage –25 to +70 °C
Air humidity	95% RH, no condensation
Ambient light	≤ 3k lux Source directly into the scanning plane
Altitude	≤ 2300 m
Limited detection zone	≤ 50 mm in front of the window cover

OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. EMC per IEC 61496-1, IEC 61000-6-2 and IEC 61000-6-3. Vibration per IEC 60068-2-6, IEC 60068-2-64, IEC 60721-3-5, IEC TR 60721-4-5 and IEC 61496-3 (5M1, sine plus random). Shock per IEC 60068-2-27, IEC 60721-3-5, IEC TR 60721-4-5 and IEC 61496-3 (5M1, single plus durable impact).

Functional safety — rated device. Unlike the rest of the OLEI 2D line, the GS1-5 carries a functional-safety type rating (see the table above), and OLEI's datasheet for it bears the TÜV Saarland "Functional Safety Approved" mark and an SGS mark. Obtain the current certificate numbers and validity dates from MorpheusTEK before design-in, and commission every protective field against the machine's own risk assessment.

Notes. Size protective fields from the 2% reflectivity figure, not the 80% measuring range, and add the 65 mm generally necessary supplement. The 50 mm zone in front of the window cover has limited detection capability and must be excluded. Ambient light is limited to ≤ 3k lux aimed directly into the scanning plane.

Installation. Upright, reverse or side mounting; optional bracket available. Mount so no structure intrudes into the 270° plane, then configure and validate every monitoring case and protective region against the machine's risk assessment before release to service. Clean the window cover on a schedule in dusty environments.

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